

CODE BLUE: EMERGENCY OPERATIONS & PATIENT FLOW ANALYTICS

Transforming emergency care data into actionable intelligence across an 11-hospital NHS network identifying pressure points, flow bottlenecks, and operational risks.

Network Snapshot:

- 11 Hospitals analyzed
- 9,994 Total Patient Visits
- 6 NHS regions covered
- 2 Hospitals in Critical pressure
- 44% Emergency admission rate
- 1.2% 4-hour breach rate



By Jeseena Parveen K

Data Analyst

Key Insights:

Nuffield and Sandwell score 73+ on hospital risk index, highest in the network and above the Critical threshold **Critical**

Severity 1 patients wait 149 min average, longer than all other severity levels, indicating a triage prioritization gap **Critical**

December overtime spikes to 4.4K hours, 44% above monthly average, signaling a recurring winter staffing crisis **High**

Critical risk patients show 27% readmission rate despite lowest ED volume, post-discharge care pathway gap identified **High**

Bristol Royal and Norfolk achieve 0% 4-hour breach, only hospitals meeting the NHS standard, representing best practice to replicate **Positive**

Recommendations:

- **Urgent operational review**
Conduct immediate review at Nuffield and Sandwell, both exceed risk score 70 with highest wait times in network. **High priority**
- **Fix triage prioritization**
Investigate why Severity 1 patients wait 149 min, lower severity patients are being seen faster across the network **High priority**
- **Winter staffing contingency**
Pre-plan additional winter capacity, December overtime consistently 44% above average across all hospitals **Medium priority**
- **Replicate best practice**
Study Bristol Royal and Norfolk protocols only two hospitals achieving 0% 4-hour breach across the network **Quick win**



Overview

Patient Flow

Staffing

Clear all slicers



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1/1/2024

12/31/2025

Region

All

Hospital

All

Department

All

9994

Total Visits

61.37

Avg Wait Time

62.50

Avg Satisfaction

54.0%

Bed Occupancy Rate

17%

Readmission Rate

0.59

Avg Burnout Index

3.3%

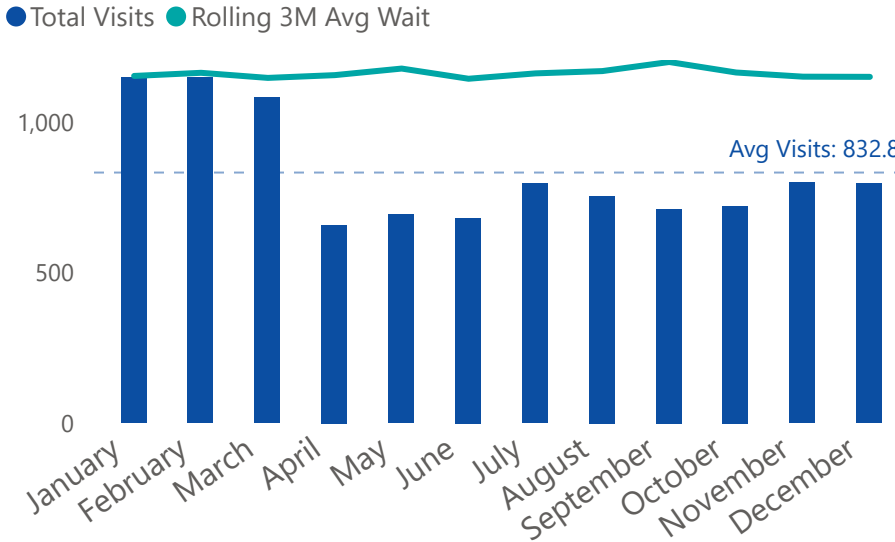
Mortality Rate

1.2%

4-Hour Breach Rate

When does demand peak?

Demand peaks in winter but bed occupancy remains consistently above 54% year-round



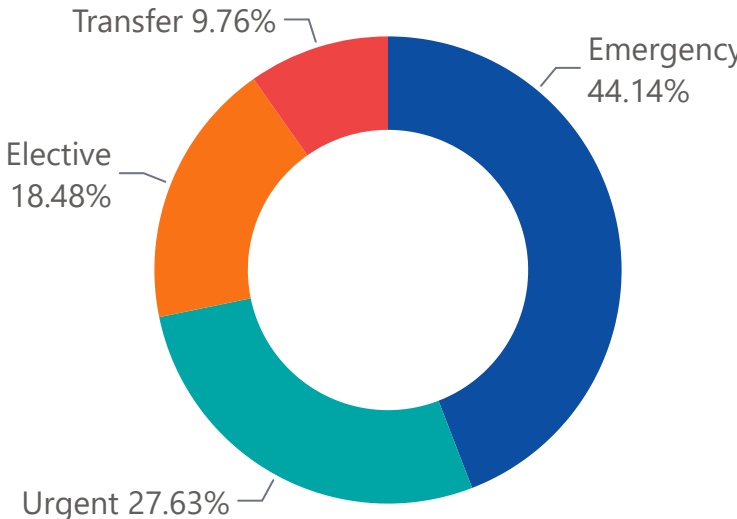
RAG
Satisfaction
Status

At Risk

RAG Wait Status

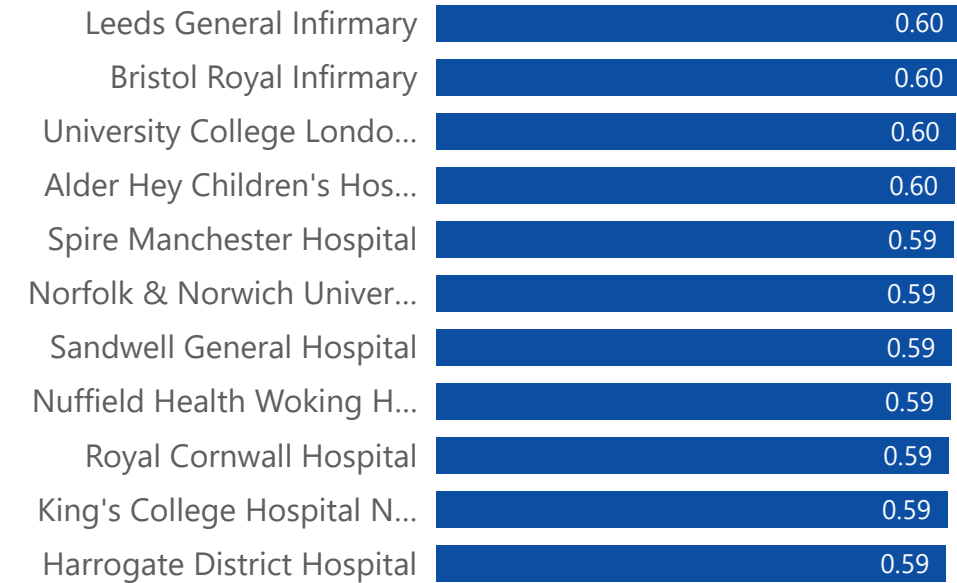
At Risk

What proportion of demand is unplanned and uncontrollable?



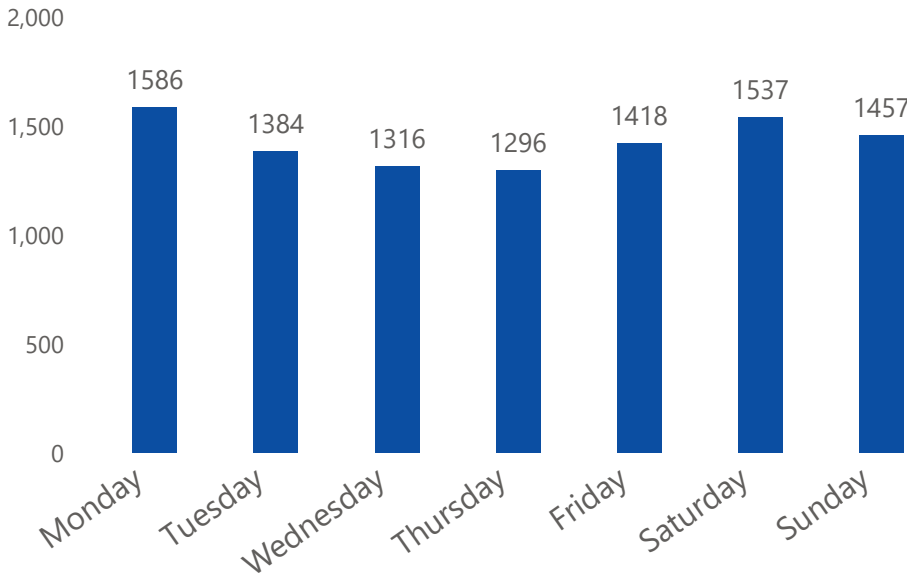
Which hospitals are burning out their staff fastest?

Burnout uniform, systemic not isolated



Which days push our emergency departments hardest?

Monday drives highest ED demand

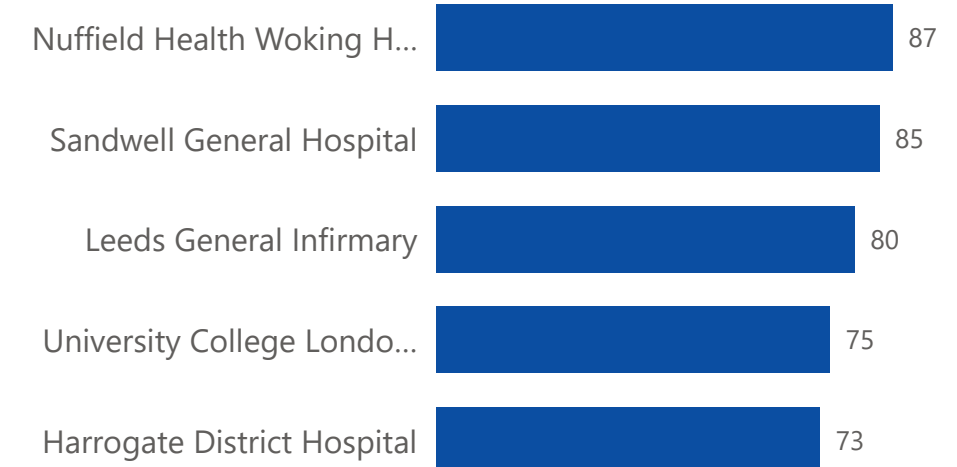


Which hospitals are closest to operational breaking point?

Hospital Name	Total Visits	Avg Wait Time	Operational Pressure Score	Pressure Category
Nuffield Health Woking Hospital	880	107.73	87.12	Critical
Sandwell General Hospital	896	103.37	84.59	Critical
Alder Hey Children's Hospital	928	62.72	72.10	High
Harrogate District Hospital	902	61.65	73.24	High
King's College Hospital NHS FT	875	28.75	60.68	High
Leeds General Infirmary	910	77.02	79.86	High
Norfolk & Norwich University Hospital	916	29.11	61.27	High
Royal Cornwall Hospital	936	48.91	66.98	High
Spire Manchester Hospital	890	48.26	65.66	High
University College London Hospitals	937	78.93	75.13	High
Bristol Royal Infirmary	924	29.77	58.96	Moderate

How many hospitals are already in the critical zone?

Nuffield and Sandwell score 73+, immediate intervention required





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1/1/2024

12/31/2025

Region

All

Hospital

All

Department

All

0.44

Emergency Admission Rate

21.4%

High Severity Rate

73.99

Avg Treatment Delay

41.49

Avg LOS Hours

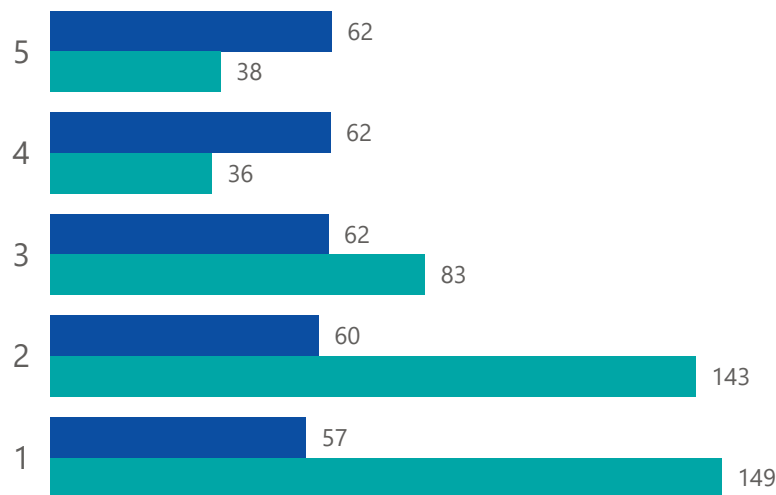
0.76

Patient Experience Index

Are the most critical patients getting seen fastest?

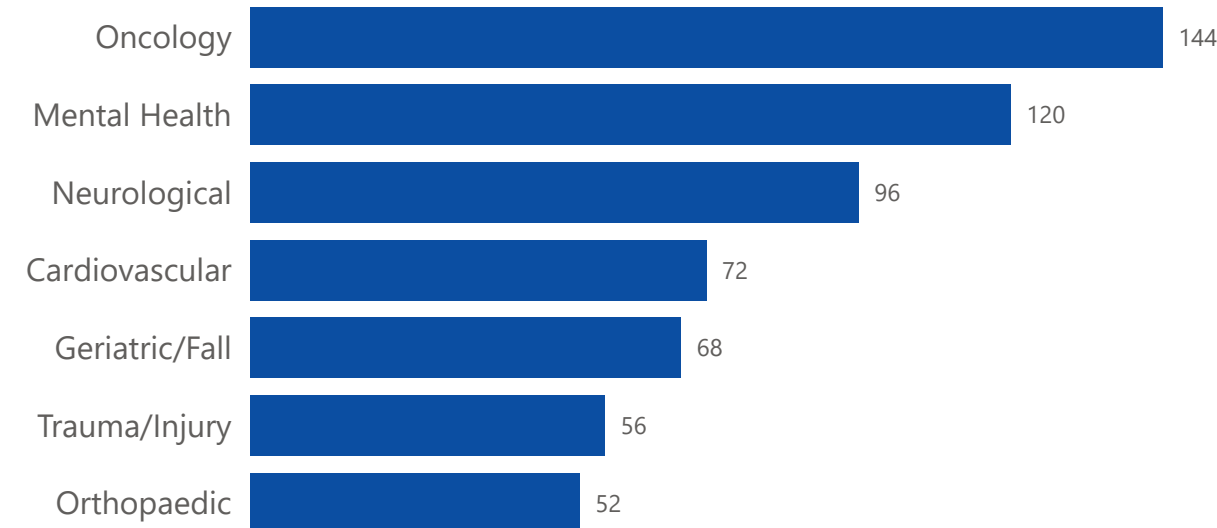
Severity 1 waits longest, triage prioritisation gap identified

● Avg Wait Time ● Avg Treatment Delay



Which diagnoses cause the longest stays?

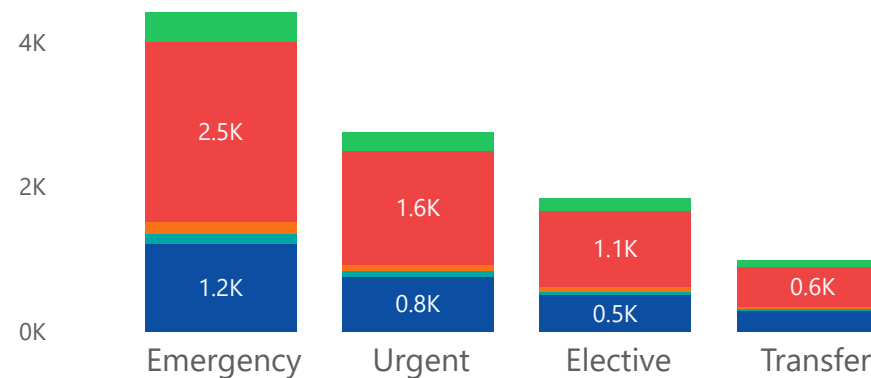
LOS uniform across categories, capacity not diagnosis drives delays



Do emergency admissions have worse outcomes than elective?

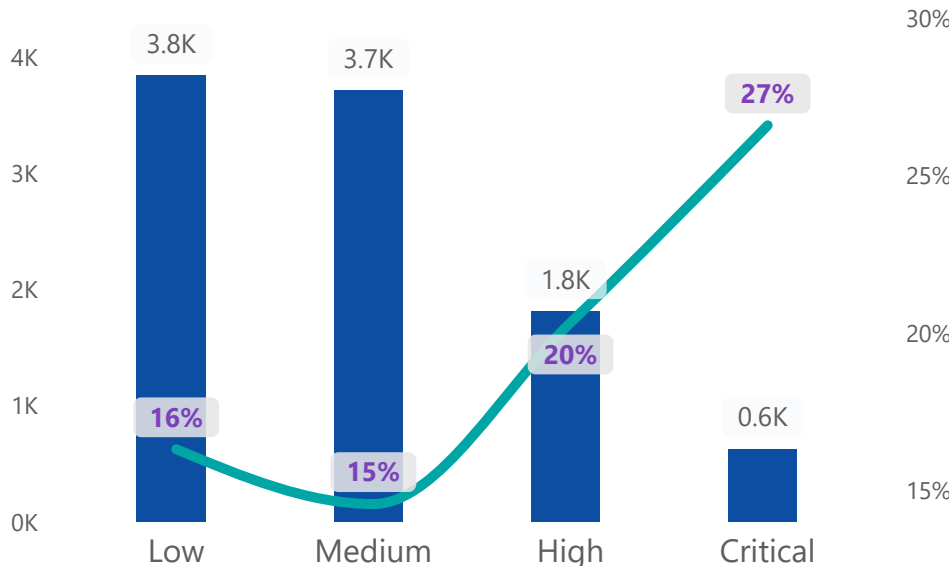
Emergency drives 44% of volume with highest Deceased proportion

● Admitted ● AMA ● Deceased ● Discharged ● Transferred



Which patient risk groups drive the most readmissions?

Critical risk patients show 27% readmission despite lowest volume

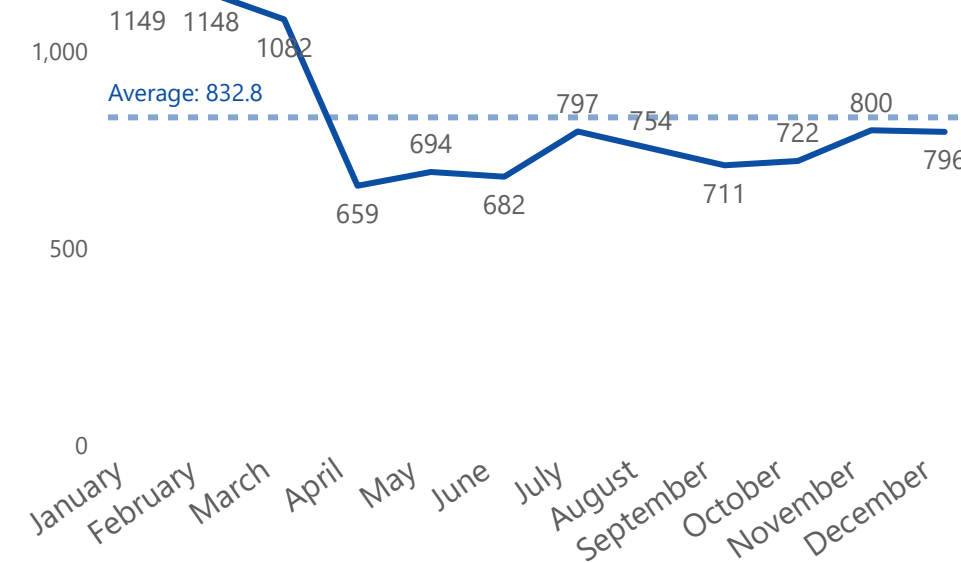


Sandwell 4.4% and Nuffield 3.6% breach rate vs 0% at Bristol and Norfolk, stark performance gap

Hospitals	Avg Wait Time	Avg LOS Hours	4-Hour Breach Rate
⊕ Sandwell General Hospital	103.37	55.67	4.4%
⊕ Nuffield Health Woking Hospital	107.73	53.74	3.6%
⊕ University College London Hospitals	78.93	40.83	1.8%
⊕ Leeds General Infirmary	77.02	41.46	1.6%
⊕ Harrogate District Hospital	61.65	62.70	0.6%
⊕ Royal Cornwall Hospital	48.91	31.39	0.5%
⊕ Spire Manchester Hospital	48.26	30.96	0.4%
⊕ King's College Hospital NHS FT	28.75	25.69	0.1%
⊕ Alder Hey Children's Hospital	62.72	64.64	0.1%
⊕ Bristol Royal Infirmary	29.77	24.65	0.0%
⊕ Norfolk & Norwich University	29.11	24.88	0.0%

How does winter and flu season spike demand?

January peak at 1,149 nearly 44% above summer baseline





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1/1/2024

12/31/2025

Region

All

Hospital

All

Department

All

0.63

Staffing Efficiency Score

8.0%

Staff Absence Rate

39.19K

Total Overtime Hours

0.17

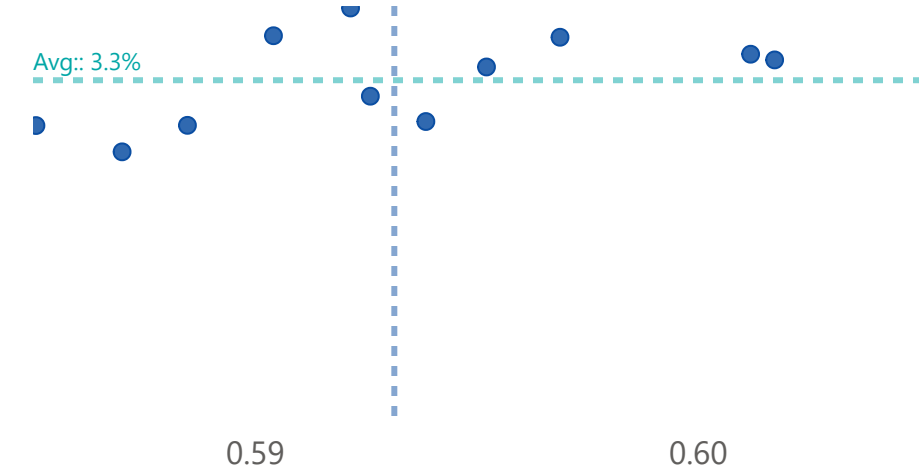
Avg Profit Margin

0.61

Hospital Risk Score

Does higher staff burnout connect to worse patient outcomes?

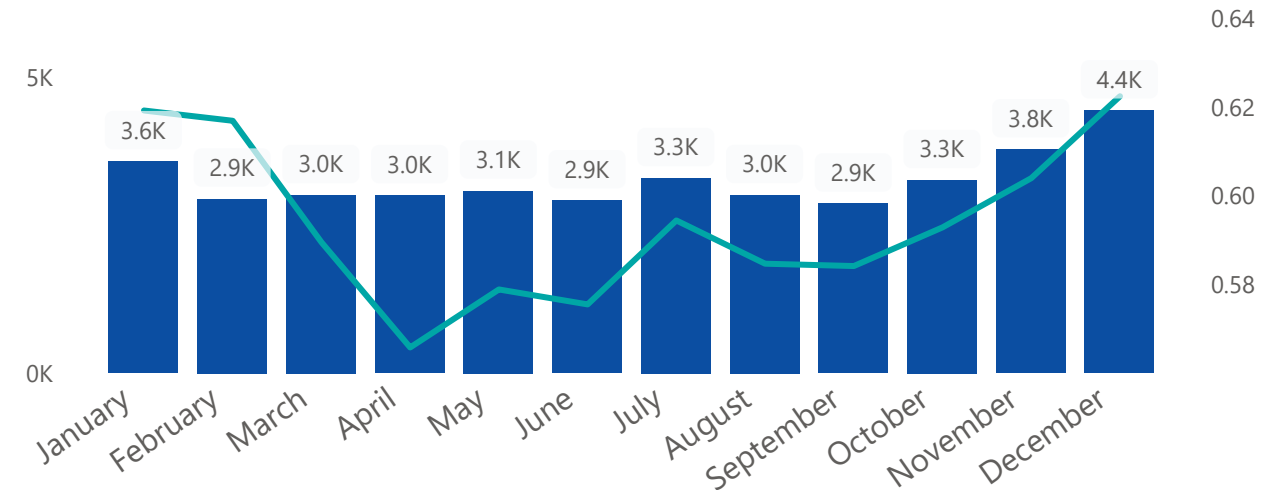
Hospitals with burnout above 0.60 show elevated mortality, staffing drives outcomes



Is overtime increasing and does it track with burnout?

December spike to 4.4K overtime hours signals winter staffing crisis

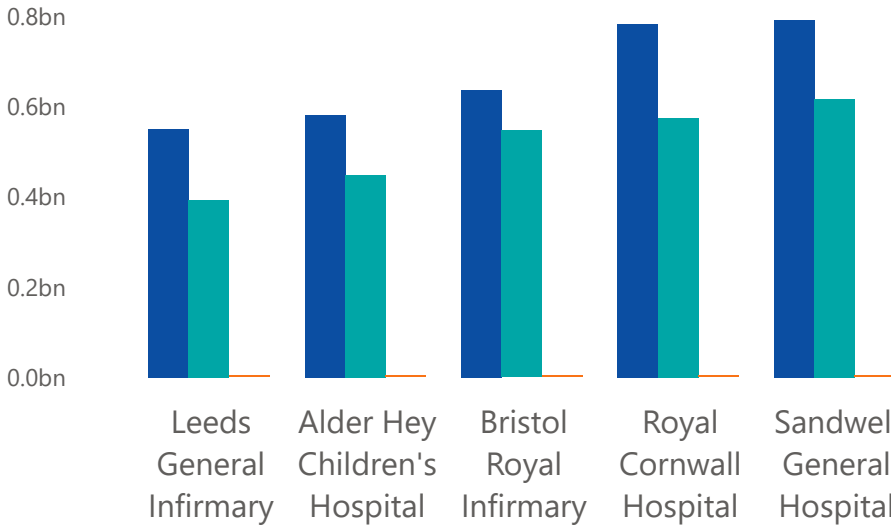
Total Overtime Hours Avg Burnout Index



Which hospitals are running at a financial loss?

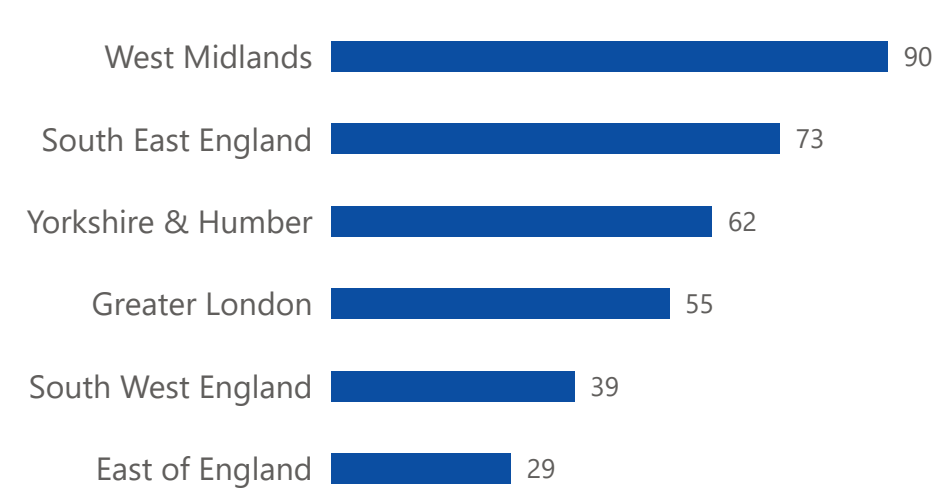
Leeds shows tightest cost-to-revenue margin

Sum of Revenue Sum of Operational_Cost Avg Profit Margin



Which regions consistently underperform across multiple metrics?

West Midlands highest wait at 90 minutes, 55% above network average



Absence rate 7.4% - 8.5% range, Norfolk highest at 8.5%

Hospitals	Avg Burnout Index	Staff Absence Rate	Total Overtime Hours	Avg Profit Margin	Patient Experience Index	Mortality Rate	Hospital Risk Score	Pressure Category
Nuffield Health Woking Hospital	0.59	7.4%	3,646.60	0.18	0.75	3.8%	0.75	Critical
Sandwell General Hospital	0.59	8.4%	3,490.20	0.16	0.76	4.0%	0.75	Critical
Alder Hey Children's Hospital	0.60	8.2%	3,804.50	0.17	0.76	3.4%	0.63	High
Harrogate District Hospital	0.59	7.9%	3,246.30	0.18	0.75	2.9%	0.59	High
King's College Hospital NHS FT	0.59	7.8%	3,319.80	0.19	0.75	2.6%	0.51	High
Leeds General Infirmary	0.60	8.4%	3,736.80	0.20	0.75	3.5%	0.66	High
Norfolk & Norwich University	0.59	8.5%	3,610.30	0.18	0.75	3.2%	0.53	High
Royal Cornwall Hospital	0.59	7.6%	3,468.50	0.16	0.75	2.9%	0.56	High
Spire Manchester Hospital	0.59	8.0%	3,468.80	0.17	0.76	2.9%	0.56	High
University College London Hospitals	0.60	8.1%	3,560.60	0.19	0.76	3.7%	0.68	High
Bristol Royal Infirmary	0.60	8.3%	3,838.20	0.14	0.76	3.6%	0.54	Moderate